

# Geotechnical Engg.

**BRANCH: CIVIL ENGG. (CE)**

COURSE CODE: CE303

## Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Geotechnical Engg. (CE303) under the Civil Engg. (CE) department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

## University Course Syllabus & Core Units

### Unit 1: Soil Properties & Classification

Three-phase soil system, void ratio, porosity, water content, unit weights, specific gravity. Index properties, consistency limits. Unified and Indian Standard Soil Classification Systems.

### Unit 2: Permeability & Seepage

Darcy's Law, coefficient of permeability, laboratory testing (constant head, falling head). Seepage analysis: flow nets, quicksand condition.

### Unit 3: Compaction & Consolidation

Compaction: Proctor test, OMC, MDD. Consolidation: Terzaghi's 1D consolidation theory, e-log p curve, compression index, settlement calculations.

### Unit 4: Shear Strength & Foundations

Mohr-Coulomb shear strength criteria. Direct shear, Triaxial compression, Unconfined compression tests. Bearing capacity: Terzaghi's theory.

*Porosity  $n = e / (1 + e)$ . Darcy's Law:  $v = k * i$ . Effective Stress:  $\sigma' = \sigma - u$ .*

## Key Formulas & Exam Pointers

CRITICAL FORMULA / PRINCIPLE:

